Submission Guidelines

Open Projects 2026

The Cultural Council, IIT Roorkee, is pleased to present **Open Projects 2026**—an initiative aimed at encouraging students to apply their skills to practical, industry-inspired problem statements across diverse domains.

The project statements have been designed to simulate real-world challenges encountered in technology, product management, consulting, design, and data-driven decision-making. Participants will have the opportunity to work on open-ended problems that require research, structured thinking, creativity, and execution.

Whether your interests lie in software development, artificial intelligence, product management, strategy, consulting, or design, Open Projects provides an opportunity to gain hands-on experience, build portfolio-worthy work, and develop skills that are highly valued during internships and placements.

Submission Guidelines

Submission Format

Participants must submit **exactly one link** as their final submission.

The submission link may be either:

- A Google Drive link containing all deliverables, or
- A GitHub repository containing all relevant files, source code, documentation, and supporting materials.

For team-based problem statements, every team member must individually submit the project through the submission form, even if all members are submitting the same project link.

Participants are responsible for ensuring that:

- The submitted link is publicly accessible.
- All required deliverables are included within the submitted link.
- Appropriate sharing permissions have been granted.
- The link remains accessible throughout the evaluation period.

Submission Limits

Participants may attempt multiple problem statements. However, **a maximum of two project submissions per student will be verified and evaluated.**

Students are encouraged to focus on quality and submit their strongest work.

Submission Portal

A Google Form will be made available for collecting submissions.

Participants will be required to provide:

- Name of Participant
- Department and Year
- Problem Statement Chosen
- Submission Link (Google Drive or GitHub Repository)

Submission Deadline

The submission portal will remain open until:

11:59 PM, 12 June 2026

Late submissions will not be considered.

AI/ML

Recommendation Systems for Personalized Content Discovery

Participation Format

This is a team problem statement with a maximum of 02 participants per team.

Teams will work with the Netflix Prize Dataset to develop a personalized recommendation system capable of learning user preferences and generating relevant content recommendations.

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1. Motivation

Every day, billions of users interact with recommendation systems that influence the movies they watch, the songs they listen to, the products they purchase, and the content they consume online.

Recommendation systems have become one of the most impactful applications of Artificial Intelligence and Machine Learning, powering personalization across platforms such as Netflix, Amazon, Spotify, YouTube, LinkedIn, and Instagram.

The ability to accurately understand user preferences and deliver relevant content directly influences user engagement, retention, and overall user experience. As a result, recommendation systems have evolved into a critical component of modern digital products.

One of the most influential milestones in the history of recommender systems was the Netflix Prize, a global machine learning competition launched to improve movie recommendation accuracy. The dataset released through this initiative remains one of the most widely studied datasets in recommendation system research and industry.

In this challenge, participants will step into the role of Machine Learning Engineers and Data Scientists tasked with building an intelligent recommendation engine capable of predicting user preferences and generating personalized content recommendations.

Dataset Overview

Dataset Link

https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data

Participants will work with the Netflix Prize Dataset, one of the most influential datasets in the history of recommender systems.

Originally released as part of the Netflix Prize competition, the dataset was created to improve Netflix's movie recommendation algorithm and has since become a benchmark dataset for recommendation system research.

The dataset contains:

- 100,480,507 movie ratings
- 480,189 users
- 17,770 movies
- Ratings on a 1–5 star scale
- Rating timestamps for each interaction
- Movie metadata including movie IDs, titles, and release years

Each rating record consists of:

- User ID
- Movie ID
- Rating
- Date of Rating

The dataset is highly sparse, meaning that each user has rated only a small fraction of the available movies. This characteristic makes it particularly suitable for studying real-world recommendation challenges such as collaborative filtering, latent factor modeling, recommendation ranking, and cold-start problems.

Participants are encouraged to use appropriate subsets of the data if computational constraints prevent training on the full dataset.

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2. Problem Statement

You are part of the machine learning team responsible for improving content discovery on a large-scale streaming platform.

Using historical user-item interaction data from the Netflix Prize Dataset, your task is to design and develop a recommendation system capable of delivering personalized content recommendations to users.

The recommendation engine should be capable of:

- Learning user preferences from historical interactions
- Predicting ratings for unseen content
- Generating personalized recommendations
- Identifying similarities between users and content items
- Improving content discovery through relevant recommendations

Participants may explore a variety of recommendation approaches, including but not limited to:

- User-Based Collaborative Filtering
- Item-Based Collaborative Filtering
- Matrix Factorization
- Singular Value Decomposition (SVD)
- Alternating Least Squares (ALS)
- Neural Collaborative Filtering
- Hybrid Recommendation Systems
- Ensemble Methods

The objective is not merely to predict ratings but to build a recommendation system that effectively captures user preferences and delivers meaningful recommendations.

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3. Mandatory and Optional Tasks

Mandatory Tasks

A. Exploratory Data Analysis

Perform a detailed analysis of the dataset and identify:

- User activity patterns
- Content popularity trends
- Rating distributions
- Data sparsity characteristics

- Interesting observations and insights

Participants should clearly communicate the business and technical implications of their findings.

B. Recommendation Model Development

Develop at least one recommendation model capable of:

- Learning user preferences
- Predicting unseen ratings
- Generating personalized recommendations

Clearly explain the methodology used and justify its suitability for the problem.

C. Model Comparison

Implement and compare at least two recommendation approaches.

Possible comparisons may include:

- User-Based vs Item-Based Collaborative Filtering
- Matrix Factorization vs Collaborative Filtering
- Traditional Methods vs Deep Learning Approaches

Participants should discuss:

- Recommendation quality
- Training complexity
- Computational efficiency
- Practical usability

D. Recommendation Generation

Generate Top-K recommendations for users and analyze the recommendations produced.

Participants should demonstrate:

- Sample recommendations
- Recommendation quality

- Success cases
- Failure cases
- Key observations

E. Evaluation

Participants must evaluate their recommendation system using the following **mandatory metrics**:

Mandatory Metrics

- **RMSE (Root Mean Squared Error)** – for measuring rating prediction accuracy.
- **MAP@10 (Mean Average Precision @ 10)** – for measuring recommendation ranking quality.

For the purpose of calculating **MAP@10**, a movie should be considered **relevant if its actual user rating is greater than or equal to 3.5**.

Participants must clearly describe:

- Their train-test split methodology
- The relevance definition used
- The procedure followed to generate Top-10 recommendations
- The methodology used to compute MAP@10

Additional Metrics (Optional)

Participants may additionally report:

- MAE
- Precision@K
- Recall@K
- NDCG
- Hit Rate
- Coverage
- Diversity Metrics

Participants must justify their evaluation strategy and discuss the trade-offs between rating prediction accuracy and recommendation ranking performance.

Optional Tasks

A. Explainable Recommendations

Provide explanations for recommendations generated by the model.

Example:

> Users who enjoyed Movie A and Movie B were also likely to enjoy Movie C.

B. Cold Start Strategy

Discuss approaches for handling:

- New users
- New content items
- Sparse interaction histories

Implementation is optional.

C. Interactive Dashboard

Build an interface that allows users to:

- View recommendations
- Explore similar content
- Analyze recommendation scores

This task is optional and will not affect eligibility for evaluation.

D. Hybrid Recommendation System

Incorporate additional metadata or external information to improve recommendation quality.

E. Deployment

Deploy the recommendation system as:

- A Web Application
- A Dashboard

- An API Service

Deployment is optional.

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4. Deliverables

Deliverable 1: Technical Report

Format: PDF

The report should include:

- Problem Understanding
- Exploratory Data Analysis
- Methodology
- Model Design
- Evaluation Metrics
- Experimental Results
- Recommendation Examples
- Key Insights
- Future Improvements

Maximum Length: 10 Pages

Deliverable 2: Source Code & GitHub Repository

The submission must include a **GitHub repository** containing:

- Data Processing Pipeline
- Model Training Pipeline
- Evaluation Scripts
- Recommendation Generation Module
- Documentation
- Instructions to reproduce results

The repository should be well-structured, documented, and reproducible.

Deliverable 3: Presentation

Format: PDF

Maximum Length: 8 Slides

The presentation should cover:

- Problem Overview
- Approach
- Key Insights
- Experimental Results
- Recommendation Examples

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5. Guidelines

1. Focus on Recommendation Quality

The objective is not simply to build a predictive model but to generate useful and relevant recommendations.

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2. Justify Your Decisions

Clearly explain:

- Model choices
- Evaluation metrics
- Feature engineering decisions
- Recommendation strategy

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3. Balance Performance and Simplicity

A simpler model with strong insights and thoughtful analysis is preferable to an unnecessarily complex solution.

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4. Encourage Experimentation

Participants are encouraged to compare multiple approaches and explore creative techniques for improving recommendation quality.

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5. Think Like a Machine Learning Engineer

Strong submissions will demonstrate not only model-building skills but also an understanding of user behavior, recommendation quality, ranking performance, and practical deployment considerations.

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6. Prioritize Reproducibility

Your work should be easy to understand, reproduce, and extend. Proper documentation and repository organization are strongly encouraged.

Evaluation Criteria

Criterion	Weightage
Data Understanding & EDA	15%
Recommendation Model Development	30%
RMSE & MAP@10 Performance	20%
Recommendation Quality & Insights	20%

Innovation & Creativity	10%
Presentation & Documentation	5%

Data-Driven Investment Intelligence Using NIFTY-50 Market Data

Participation Format

This is a team problem statement.

Teams may consist of up to 02 participants.

Participants are expected to collaboratively design and develop an AI-powered investment intelligence platform using the provided NIFTY-50 dataset. All submitted work must be original and completed within the competition timeline.

The challenge evaluates not only machine learning and analytical capabilities but also the ability to transform financial data into practical decision-support systems that can assist investors in making informed decisions.

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1. Motivation

Financial markets generate massive amounts of data every day, making it increasingly difficult for investors to identify meaningful patterns and make informed decisions.

While stock market data is publicly available, transforming raw historical prices and trading volumes into actionable insights requires sophisticated analytical techniques.

The NIFTY-50 index represents fifty of the largest and most influential companies listed on the National Stock Exchange of India, spanning multiple sectors such as Banking, Information Technology, Energy, Consumer Goods, Pharmaceuticals, Financial Services, and Manufacturing.

This challenge invites participants to build an AI-powered investment intelligence platform capable of extracting insights, identifying opportunities, assessing risks, and supporting data-driven investment decisions using historical market data.

Unlike traditional stock-price prediction competitions, the focus of this challenge is on creating practical decision-support systems that combine machine learning, financial analytics, explainability, and user-centric design.

Dataset Overview

Dataset Link:
https://www.kaggle.com/datasets/rohanrao/nifty50-stock-market-data/data

Additional datasets :

https://www.kaggle.com/datasets/stoicstatic/india-stock-data-nse-1990-2020

(Students are requested to use only the provided datasets)

Participants will work with the **NIFTY-50 Stock Market Dataset**, which contains historical market data for companies that have been part of the NIFTY-50 index of the National Stock Exchange (NSE) of India.

The dataset spans from **January 2000 to April 2021** and covers companies across multiple sectors including Banking, Information Technology, Energy, Consumer Goods, Pharmaceuticals, Financial Services, and Manufacturing.

The dataset consists of:

Historical Stock Data

Daily records for each company containing:

- Open Price
- High Price
- Low Price
- Close Price
- Traded Volume
- Turnover

Company Metadata

Additional information including:

- Company Name
- Sector

- Industry Classification

The dataset provides more than two decades of historical market activity, enabling participants to analyze long-term trends, market cycles, sectoral performance, volatility patterns, and investment opportunities.

Examples of derived indicators include:

- Moving Averages (MA)
- Exponential Moving Averages (EMA)
- Relative Strength Index (RSI)
- MACD
- Bollinger Bands
- Volatility Indicators
- Momentum Indicators

The challenge is designed to evaluate a participant's ability to transform raw historical market data into actionable investment intelligence and decision-support tools.

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2. Problem Statement

You are part of the Data Science and Investment Analytics team of a financial technology company.

Your objective is to develop an intelligent investment intelligence platform capable of transforming historical market data into meaningful insights that can assist investors in making informed decisions.

The platform should help users:

- Analyze historical stock performance.
- Evaluate investment opportunities.
- Assess portfolio risk.
- Understand market trends and behavior.
- Generate actionable investment insights.
- Support data-driven decision making.

Participants are free to use any combination of:

- Machine Learning
- Time Series Forecasting
- Statistical Modeling

- Deep Learning
- Portfolio Optimization
- Financial Analytics

The final solution should go beyond simple stock-price forecasting and focus on delivering practical investment intelligence.

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3. Mandatory and Optional Tasks

Mandatory Tasks

A. Stock Predictor Engine

Develop predictive models capable of forecasting future stock behavior using historical market data.

Examples:

- Predict future stock prices or returns over a specified investment horizon.
- Predict the direction of stock price movement (upward or downward).
- Forecasts should be supported by appropriate model evaluation and validation techniques.

Evaluation Metrics:
Participants may use metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 Score, and Directional Accuracy to assess model performance. Additional evaluation metrics may be incorporated if appropriately justified.

B. Portfolio Construction Module

Generate investment portfolios for different investor profiles.

Examples:

- Conservative Investor
- Balanced Investor
- Aggressive Investor

The system should recommend portfolio allocations and justify its decisions.

C. Risk Assessment Module

Evaluate the historical risk associated with individual stocks or portfolios.

Possible Metrics Include:

- Volatility
- Sharpe Ratio
- Sortino Ratio
- Maximum Drawdown
- Risk-Adjusted Return

Participants may explore additional risk metrics where appropriate.

Optional Tasks

A. Explainable AI Framework

Provide explanations supporting recommendations and insights generated by the system.

B. Personalized Investment Strategies

Design investment recommendations tailored to different risk tolerances and investment goals.

C. Market Anomaly Detection

Identify unusual patterns or events within historical market data.

Examples:

- Sudden Volatility Spikes
- Extreme Drawdowns
- Unusual Trading Activity

D. Forecasting Module

Develop predictive models for:

- Future Price Trends
- Volatility Estimation
- Return Forecasting

Participants should clearly discuss assumptions and limitations.

E. Deployment

Deploy the solution as:

- A Web Application
- An Interactive Dashboard
- A Cloud-Based Service
- A Local Desktop Application

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4. Constraints

Participants may use only the datasets provided by the organizers.

Allowed

- Feature Engineering
- Statistical Transformations
- Technical Indicators
- Machine Learning Models
- Deep Learning Models
- Portfolio Optimization Techniques

Examples include:

- Moving Averages
- Exponential Moving Averages
- RSI
- MACD
- Bollinger Bands
- Momentum Indicators
- Volatility Indicators

Not Allowed

- Live Market Data
- Financial APIs
- News Datasets
- Social Media Sentiment Data

- Proprietary Financial Data
- Alternative Market Datasets

The objective is to evaluate how effectively participants can leverage the provided dataset.

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5. Deliverables

Deliverable 1: Working Prototype

A functional application demonstrating the implemented capabilities.

Deliverable 2: Technical Report

Format: PDF

The report should include:

- Exploratory Data Analysis (EDA)
- Feature Engineering
- Methodology
- Model Architecture
- Portfolio Construction Logic
- Risk Assessment Methodology
- Explainability Techniques
- Key Insights

Maximum Length: 12 Pages

Deliverable 3: Public GitHub Repository

The repository should contain:

- Source Code
- Model Files
- Configuration Files
- Documentation

The solution should be reproducible.

Deliverable 4: README.md

Include instructions for:

- Environment Setup
- Dependency Installation
- Running the Application
- Reproducing Results

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6. Guidelines

1. Focus on Decision Support

The objective is not simply to predict stock prices but to assist users in making informed investment decisions.

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2. Justify Your Recommendations

All investment recommendations and portfolio suggestions should be supported by quantitative evidence.

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3. Emphasize Explainability

Users should be able to understand why a recommendation or insight was generated.

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4. Balance Complexity and Usability

Sophisticated analytics should be presented in a manner that remains understandable and actionable.

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5. Encourage Innovation

Participants are encouraged to explore novel approaches for investment intelligence, risk assessment, portfolio optimization, and explainable AI.

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7. Evaluation Criteria

Criterion	Weightage
Quality of Investment Insights	25%
Technical Innovation	20%
Portfolio & Risk Analytics	20%
Explainability & Transparency	20%
Reproducibility & Documentation	15%

Software Development

Smart Asset Management and Resource Allocation Platform

Participation Format

This is a team problem statement.

Teams may consist of up to the maximum number of 03 participants.

Participants are expected to design and develop a full-stack asset management platform that enables efficient inventory tracking, resource allocation, and operational visibility.

The challenge evaluates software engineering, database design, system architecture, API development, and user experience design skills.

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1. Motivation

Organizations of all sizes rely on efficient asset management systems to track inventory, allocate resources, monitor utilization, and maintain accountability.

From universities and event management companies to logistics providers and enterprises, poor inventory visibility often results in scheduling conflicts, resource underutilization, delayed returns, and operational inefficiencies.

The Cultural Council of IIT Roorkee faces a similar challenge while managing a large pool of shared resources such as:

- DSLR Cameras
- Studio Lighting Equipment
- Audio Systems
- Costumes
- Stage Props
- Recording Equipment
- Event Infrastructure

These assets are frequently shared across multiple sections and events, making efficient resource allocation critical.

Currently, resource management often relies on fragmented communication channels, spreadsheets, manual registers, and informal coordination processes.

This challenge provides an opportunity to build a real-world asset management platform that mirrors problems commonly encountered in inventory management, resource planning, logistics operations, and enterprise software systems.

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2. Problem Statement

Design and develop a full-stack Asset Management and Resource Allocation Platform that enables organizations to efficiently manage shared resources.

The platform should provide a centralized system for:

- Inventory management
- Asset booking
- Approval workflows
- Resource allocation
- Utilization tracking
- Operational analytics

The system should support role-based workflows for administrators and resource consumers while maintaining simplicity, usability, and scalability.

Participants should focus on building a robust and user-friendly solution capable of handling real-world asset management scenarios.

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3. Functional Requirements

Mandatory Features

A. User Authentication

Implement a secure authentication system.

Users should be able to:

- Register and log in
- Access role-specific functionality
- Maintain secure sessions

Participants may use:

- JWT Authentication
- Session-Based Authentication
- OAuth (Optional)

B. Inventory Management

Administrators should be able to:

- Add assets
- Edit asset information
- Delete assets
- Categorize assets
- Manage available quantities

Each asset should contain information such as:

- Asset Name
- Category
- Description
- Quantity Available
- Status

C. Asset Discovery and Booking

Users should be able to:

- Browse available assets
- Search and filter inventory
- View asset availability
- Request assets for specific durations

The platform should prevent booking requests exceeding available inventory.

D. Approval Workflow

Administrators should be able to:

- Review booking requests
- Approve requests
- Reject requests
- View active allocations

Users should be able to track the status of their requests.

E. Asset Issue and Return Management

The platform should support:

- Asset issuance
- Asset return tracking
- Due date management
- Asset status updates

The system should maintain accurate inventory counts throughout the asset lifecycle.

F. Analytics Dashboard

Build a dashboard providing insights such as:

- Most frequently utilized assets
- Asset utilization rates
- Active bookings
- Available inventory
- Overdue returns

Visualizations may include:

- Bar Charts
- Pie Charts
- Line Graphs
- Summary Cards

G. Borrowing History

Users should be able to:

- View their borrowing history
- Track active bookings
- Monitor past requests

Administrators should be able to view system-wide activity.

Optional Features

A. Notification System

Notify users regarding:

- Booking approvals
- Booking rejections
- Upcoming return deadlines
- Overdue assets

Notifications may be delivered through:

- Email
- In-App Notifications

B. Audit Logs

Track important actions such as:

- Asset creation
- Inventory updates
- Booking approvals
- Asset returns

C. QR Code-Based Asset Operations

Allow administrators to:

- Generate QR codes for assets
- Scan assets during issue and return

- View asset information instantly

D. Asset Health Tracking

Track:

- Asset condition
- Maintenance history
- Damage reports

E. Dockerized Deployment

Provide a containerized development environment using Docker and Docker Compose.

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4. Deliverables

Deliverable 1: Working Application

A functional application demonstrating the implemented features.

Deliverable 2: Design Document

Format: PDF

The document should include:

- Problem Understanding
- System Architecture
- Database Schema
- Entity Relationship Diagram (ERD)
- API Overview
- Design Decisions

Maximum Length: 8 Pages

Deliverable 3: Public GitHub Repository

The repository should contain:

- Source Code
- Configuration Files
- Database Scripts
- Documentation

The project should be reproducible.

Deliverable 4: README.md

The README should include:

- Project Overview
- Technology Stack
- Setup Instructions
- Running the Application
- Feature List

Deliverable 5: Demonstration Video

Duration: Maximum 3 Minutes

The video should demonstrate:

- User Authentication
- Asset Booking Workflow
- Approval Process
- Dashboard Functionality
- Key Technical Features

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5. Guidelines

1. Prioritize Core Functionality

A well-executed booking and inventory management system is preferable to an incomplete feature-heavy platform.

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2. Focus on User Experience

The platform should be intuitive and easy to use for both administrators and users.

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3. Maintain Data Integrity

Inventory counts, bookings, and returns should remain consistent throughout the system.

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4. Design for Scalability

Consider how the system would support larger inventories and increased user activity.

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5. Follow Good Engineering Practices

Strong submissions should demonstrate:

- Clean Code
- Modular Design
- Proper Documentation
- Maintainability

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6. Evaluation Criteria

Criterion	Weightage
Functionality & Feature Completeness	30%
System Design & Architecture	20%

Database Design & Backend Logic	15%
User Experience & Interface Design	15%
Code Quality & Documentation	10%
Innovation & Additional Features	10%

Bonus Considerations

Additional credit may be awarded for:

- Notification Systems
- QR-Based Operations
- Audit Logging
- Asset Health Tracking
- Dockerized Deployment
- Advanced Analytics

Real-Time Campus Mobility and Ride Management Platform

Participation Format

This is a team problem statement.

Teams may consist of up to the maximum number of 04 participants.

Participants are expected to design and develop a real-time ride management platform that enables passengers and drivers to seamlessly connect within a campus environment.

The challenge evaluates full-stack development, real-time communication systems, database design, API development, state management, and user experience design skills.

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1. Motivation

Efficient transportation is a critical component of modern campuses, universities, airports, hospitals, logistics hubs, and smart cities.

Users require quick and reliable transportation, while drivers need visibility into ride demand and passenger requests. Without centralized coordination, transportation systems often suffer from inefficient resource utilization, delays, uneven demand distribution, and poor user experiences.

The IIT Roorkee campus spans a large geographical area and relies heavily on e-rickshaws for last-mile transportation. However, ride requests and driver availability are currently coordinated through fragmented and informal mechanisms.

This challenge provides an opportunity to build a real-world mobility platform that mirrors many of the engineering challenges encountered in ride-hailing and dispatch systems, including:

- Real-time communication
- Geospatial data handling
- Ride assignment workflows
- State synchronization

- Dashboard analytics
- Multi-user coordination

Participants will develop a campus-scale ride management platform capable of connecting passengers and drivers through a centralized digital system.

---

2. Problem Statement

Design and develop a responsive web or mobile application that enables passengers and drivers to discover, request, assign, and manage rides within a campus environment.

The platform should provide:

- User authentication
- Driver onboarding
- Ride request management
- Ride assignment workflows
- Real-time ride updates
- Driver analytics and insights

The system should prioritize usability, reliability, and responsiveness while remaining scalable and maintainable.

Participants are encouraged to focus on solving the core ride-management problem effectively rather than attempting to replicate every feature of commercial ride-hailing platforms.

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3. Functional Requirements

Mandatory Features

A. User Authentication & Profile Management

Implement a secure authentication system.

Passenger Accounts

- Registration
- Login

- Profile Management

Driver Accounts

- Registration
- Login
- Vehicle Information
- Driver Verification Information

Participants may implement:

- JWT Authentication
- Session-Based Authentication
- OAuth Authentication (Optional)

B. Driver Availability Management

Drivers should be able to:

- Go Online
- Go Offline
- Update current availability status

Passengers should be able to view available drivers before requesting rides.

C. Ride Request Workflow

Passengers should be able to:

- Request a ride
- Specify pickup location
- Specify destination
- View ride status

Drivers should be able to:

- View incoming ride requests
- Accept requests
- Reject requests

The system must ensure that a ride can only be assigned to a single driver.

D. Real-Time Ride Updates

Implement real-time communication using technologies such as:

- WebSockets
- Socket.IO
- Server-Sent Events

The platform should support:

- Live ride status updates
- Driver availability updates
- Ride assignment notifications

This is the core engineering component of the challenge.

E. Ride Lifecycle Management

Support complete ride tracking through states such as:

- Requested
- Accepted
- In Progress
- Completed
- Cancelled

The system should maintain a consistent ride state throughout the workflow.

F. Driver Dashboard

Provide a dashboard displaying:

- Total rides completed
- Active rides
- Ride history
- Ratings received
- Basic ride statistics

Visualizations may include:

- Summary Cards
- Charts

- Activity Tables

G. Ratings & Feedback

Passengers should be able to:

- Rate completed rides
- Provide optional written feedback

The platform should maintain:

- Average driver ratings
- Feedback history
- Driver performance summaries

Optional Features

A. Live Map Integration

Display:

- Driver locations
- Pickup points
- Active rides

using mapping solutions such as:

- Leaflet
- OpenStreetMap
- Google Maps

B. Ride Scheduling

Allow users to:

- Schedule rides for future time slots
- Manage upcoming bookings

Examples:

- Early morning classes
- Club rehearsals
- Event transportation

C. Digital Payments

Simulate or integrate:

- UPI Payments
- QR Code Payments
- Payment History

D. Demand Analytics

Analyze historical ride data and identify:

- Peak demand hours
- Popular pickup locations
- Ride demand patterns

E. Demand Forecasting

Use machine learning techniques to predict:

- High-demand periods
- Potential ride hotspots

Examples:

- Linear Regression
- Decision Trees
- Time-Series Models

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4. Deliverables

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Deliverable 3: Public GitHub Repository

The repository should contain:

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The solution should be reproducible.

Deliverable 4: README.md

The README should include:

- Project Overview
- Technology Stack
- Setup Instructions
- Running the Application
- Feature List

Deliverable 5: Demonstration Video

Duration: Maximum 3 Minutes

The video should demonstrate:

- User Registration & Login
- Ride Request Workflow
- Ride Assignment Process
- Real-Time Updates
- Driver Dashboard
- Ratings & Feedback System

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5. Guidelines

1. Prioritize Core Ride Workflows

A reliable ride management system is preferable to numerous incomplete features.

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2. Focus on Real-Time Communication

The strongest submissions will effectively utilize WebSockets or similar technologies to deliver live updates.

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3. Maintain Data Consistency

Ride states, assignments, and driver availability should remain synchronized across the platform.

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4. Design for Scalability

Consider how the system would support larger numbers of users and ride requests.

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5. Follow Good Engineering Practices

Strong submissions should demonstrate:

- Clean Code
- Modular Architecture
- Proper Documentation
- Maintainability

---

6. Evaluation Criteria

Criterion	Weightage
Functionality & Feature Completeness	30%
Real-Time System Design	20%
Backend & Database Design	15%
User Experience & Interface Design	15%
Code Quality & Documentation	10%
Innovation & Additional Features	10%

Bonus Considerations

Additional credit may be awarded for:

- Live Map Integration
- Ride Scheduling
- Digital Payments
- Demand Analytics
- Demand Forecasting
- Advanced Dashboard Features

Product Management

Product Strategy for an Opportunity Discovery Platform

Participation Format

This is an individual problem statement.

Participants are required to work and submit individually. All research, analysis, product strategy, prototypes, presentations, and supporting deliverables must be prepared by a single participant.

The challenge evaluates product thinking, user research, prioritization, marketplace design, growth strategy, recommendation systems thinking, and the ability to design products that solve meaningful user problems.

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1. Motivation

Modern communities generate an overwhelming number of opportunities, events, programs, workshops, competitions, recruitments, internships, fellowships, and initiatives.

As communities grow larger and more diverse, discovering relevant opportunities becomes increasingly difficult. The challenge is often not a lack of opportunities, but rather an inability to connect the right opportunity with the right user at the right time.

Students today rely on a fragmented combination of messaging groups, social media posts, emails, posters, and word-of-mouth communication. As a result:

- Students frequently miss opportunities aligned with their interests and goals.
- New members experience significant information overload.
- Organizations struggle to reach relevant audiences.
- Participation decisions are often driven by visibility rather than genuine interest.
- Engagement remains concentrated among a small subset of users.

These challenges are not unique to educational institutions. Similar discovery problems exist across professional communities, creator ecosystems, learning platforms, networking products, and career marketplaces.

The opportunity lies in designing a platform that can intelligently match users with relevant opportunities while improving discoverability, engagement, and decision-making.

---

2. Problem Statement

You have been appointed as the Product Manager responsible for conceptualizing an Opportunity Discovery Platform.

The platform aims to help users discover opportunities that align with their interests, goals, skills, preferences, and availability, while enabling organizations to effectively reach relevant audiences.

Your responsibility is not to build the product.

Instead, your task is to determine:

- What problem the platform should solve.
- Who the primary users should be.
- What features should be prioritized.
- How value should be created for both users and organizations.
- How the platform should achieve adoption and retention.
- How recommendations should be surfaced.
- How success should be measured.

The platform should help users:

- Discover relevant opportunities.
- Explore events, workshops, competitions, recruitments, internships, and initiatives.
- Make informed participation decisions.
- Manage involvement across multiple activities.

The platform should help organizations:

- Reach relevant audiences.
- Improve participation and engagement.
- Reduce dependence on fragmented communication channels.
- Understand audience behavior and interests.

Your final solution should demonstrate strong product judgment, prioritization, user-centric thinking, and a deep understanding of marketplace dynamics.

---

3. Mandatory and Optional Tasks

Mandatory Tasks

A. User Research & Problem Discovery

Identify and analyze the needs of key stakeholders.

At a minimum, consider:

- First-Year Students
- Senior Undergraduate Students
- Postgraduate Students
- Student Leaders
- Club and Society Representatives
- Event Organizers

Clearly identify:

- User pain points
- Existing alternatives
- Behavioral patterns
- Unmet needs
- User motivations

Participants should identify the primary target user and justify their choice.

B. Product Requirements Document (PRD)

Create a comprehensive PRD covering:

- Problem Definition
- Product Vision
- User Personas
- User Journey Maps
- Core Value Proposition

- Key Features
- MVP Scope
- Prioritization Framework
- Success Metrics

The PRD should demonstrate clear product reasoning and structured thinking.

C. MVP Definition & Prioritization

Define the Minimum Viable Product (MVP).

Clearly justify:

- Features included in Version 1
- Features excluded from Version 1
- Product trade-offs
- Prioritization decisions

Participants are encouraged to use frameworks such as:

- RICE
- MoSCoW
- Kano Model
- Impact vs Effort

The emphasis should be on solving the core problem with the smallest viable product.

D. Product Prototype

Design a high-fidelity prototype demonstrating key user journeys.

Suggested workflows include:

- User Onboarding
- Interest Selection
- Opportunity Discovery
- Personalized Recommendations
- Opportunity Details Page
- Registration Workflow
- Organization Publishing Workflow

The prototype should clearly communicate the intended user experience.

E. Growth & Adoption Strategy

Design a strategy addressing:

User Acquisition

How will users discover and adopt the platform?

Organization Onboarding

How will opportunity providers be encouraged to participate?

Retention

Why will users continue returning to the platform?

Engagement

How will the platform remain useful over time?

Participants should consider both sides of the ecosystem and address potential cold-start challenges.

F. Success Metrics Framework

Define the metrics that determine whether the product is successful.

Examples include:

- Monthly Active Users (MAU)
- Weekly Active Users (WAU)
- User Retention
- Opportunity Discovery Rate
- Registration Conversion Rate
- Time-to-Discovery
- Engagement Rate

Participants should identify:

- North Star Metric
- Supporting Metrics
- Leading Indicators
- Lagging Indicators

Clearly justify why these metrics matter.

Optional Tasks

A. Recommendation Strategy

Design a conceptual recommendation framework.

Recommendations may be generated using:

- User Interests
- Past Interactions
- Peer Behavior
- Event Metadata
- User Goals
- Community Participation Patterns

Technical implementation is not required.

The focus should be on product thinking and user value.

B. Marketplace Strategy

The platform serves both users and opportunity providers.

Design a strategy for:

- Solving the cold-start problem
- Balancing supply and demand
- Ensuring content quality
- Maintaining platform trust

C. Gamification Strategy

Design mechanisms that encourage:

- Exploration
- Participation
- Consistent Engagement

- Community Contributions

Examples may include:

- Achievement Systems
- Streaks
- Participation Badges
- Reputation Systems

D. Product Roadmap

Develop a roadmap for the first 12 months after launch.

Include:

- Feature Expansion
- Growth Priorities
- Product Milestones
- Strategic Bets

E. Analytics Dashboard

Design dashboards that help organizations understand:

- Reach
- Engagement
- Conversion
- Audience Segmentation
- Opportunity Performance

---

4. Deliverables

Deliverable 1: Product Requirements Document (PRD)

Format: PDF

The PRD should include:

- Problem Analysis
- User Personas
- User Journeys
- Product Vision
- MVP Scope
- Prioritization Framework
- Growth Strategy
- Success Metrics

Deliverable 2: Interactive Prototype

Format:

- Figma
- Framer
- Adobe XD
- Any Equivalent Design Tool

The prototype should showcase the primary user journeys and overall product experience.

Deliverable 3: Product Strategy Deck

Format: PDF

Maximum Length: 10 Slides

The deck should communicate:

- Problem
- User Insights
- Product Vision
- Proposed Solution
- MVP Scope
- Growth Strategy
- Success Metrics

The deck should be concise, structured, and executive-friendly.

---

5. Guidelines

1. Think Like a Product Manager

Focus on solving the right problem rather than proposing the largest number of features.

---

2. Prioritize Ruthlessly

A focused MVP is preferable to a feature-heavy product.

---

3. Support Decisions with Reasoning

Every major product decision should be justified through user needs, expected impact, strategic considerations, or product principles.

---

4. State Assumptions Clearly

Reasonable assumptions may be made where necessary. All assumptions should be explicitly stated.

---

5. Design for Adoption

A product creates value only if users and organizations actively choose to use it.

---

6. Focus on User Outcomes

Success should be measured through meaningful user outcomes rather than feature count or vanity metrics.

---

7. Balance User and Organization Needs

Strong products create value for both sides of the ecosystem.

Participants should demonstrate how their solution balances user experience with organizational objectives.

---

6. Evaluation Criteria

Criterion	Weightage
User Understanding & Research	25%
Product Thinking & Problem Solving	25%
Prioritization & MVP Design	20%
Growth & Adoption Strategy	15%
Communication & Presentation	15%

A Data-Driven Strategy for Revitalizing IIT Roorkee's Performance at the Inter IIT Cultural Meet

Participation Format

This is an individual problem statement.

Participants are required to work and submit individually. All research, analysis, product strategy, prototypes, presentations, and supporting deliverables must be prepared by a single participant.

---

1. Motivation

The Inter IIT Cultural Meet is one of the most prestigious cultural competitions in the country, bringing together talented contingents from IITs across a diverse range of disciplines including dramatics, dance, music, literary arts, design, fine arts, filmmaking, photography, and quizzes.

For IIT Roorkee, the meet represents far more than a competition. It is an opportunity to showcase the institute's creativity, cultural vibrancy, and artistic excellence on a national stage.

IIT Roorkee has a rich cultural legacy and has consistently produced exceptional performers across disciplines. However, recent results indicate a concerning trend. After securing an impressive **3rd position overall at the Inter IIT Cultural Meet in 2023**, the institute's standing has gradually declined, culminating in a **7th-place finish in 2025**, with performance also dipping in the intervening year.

While rankings alone do not tell the complete story, this consistent decline raises important questions about the institute's cultural ecosystem.

- Is talent being discovered early enough?
- Are students receiving the right mentorship and training?
- Are cultural sections able to retain promising performers?
- Are resources being allocated effectively?
- Are participation opportunities accessible to a wider student population?
- Is there sufficient data to guide decision-making?
- Are there systemic challenges that remain unidentified?

The Cultural Council believes that reversing this trend requires more than short-term interventions or last-minute preparation before competitions. It requires a sustainable, data-driven, and scalable approach to identifying, nurturing, and supporting cultural talent across the institute.

To address this challenge, the Cultural Council is seeking innovative solutions that can strengthen IIT Roorkee's cultural ecosystem and create sustained improvements in performance at future editions of the Inter IIT Cultural Meet.

---

2. Problem Statement

The Cultural Council seeks to develop a long-term strategy that can help IIT Roorkee strengthen its performance at the Inter IIT Cultural Meet and re-establish itself among the top-performing contingents.

You have been appointed as a Product Manager to investigate this challenge and propose a solution.

Your task is not to organize the next contingent.

Your task is to identify the root causes behind the current performance trend and design a product, platform, system, process, program, or initiative that can create measurable and sustained improvements over the coming years.

Your solution should focus on building capabilities rather than delivering one-time interventions.

The proposed solution may address one or more aspects of the cultural ecosystem, including but not limited to:

- Talent discovery and recruitment
- Training and mentorship
- Participant retention
- Performance tracking
- Community engagement
- Resource allocation
- Event participation
- Knowledge transfer between batches
- Alumni involvement
- Cross-section collaboration
- Data-driven decision making

Participants are encouraged to think beyond technology products. The best solution may be a platform, a program, a framework, a decision-support system, or a combination of multiple interventions.

The objective is to create a sustainable mechanism that strengthens IIT Roorkee's cultural ecosystem and improves its competitiveness in future editions of the Inter IIT Cultural Meet.

---

3. Mandatory and Optional Tasks

Mandatory Tasks

A. Problem Discovery and Root Cause Analysis

Before proposing a solution, identify the factors that may be contributing to the decline in performance.

Your analysis should include:

- Existing challenges
- Stakeholder pain points
- Potential bottlenecks
- Opportunity areas
- Key assumptions

Participants are expected to support their analysis through research, observations, interviews, surveys, publicly available information, or reasonable assumptions.

B. Stakeholder Analysis

Identify and analyze the needs of key stakeholders, including:

- Students
- Cultural Sections
- Section Secretaries
- Cultural Council
- Coaches and Mentors
- Alumni
- Inter IIT Contingent Members

Participants may consider additional stakeholders if relevant.

C. Product Strategy

Define:

- Vision
- Target users
- Problem statement
- Value proposition
- Strategic objectives

Clearly explain why your proposed intervention is the most effective approach to solving the identified problem.

D. MVP Definition

Design a Minimum Viable Product (MVP) or Minimum Viable Initiative (MVI).

Clearly identify:

- Core features or components
- Prioritization rationale
- Scope exclusions
- Expected impact

Participants should focus on realistic and implementable solutions rather than attempting to solve every problem simultaneously.

E. Success Framework

Define measurable outcomes for your solution.

Examples include:

- Talent discovery rate
- Recruitment conversion rate
- Participant retention rate
- Training engagement levels
- Section-level participation growth
- Performance improvement indicators
- Medal conversion rate
- Overall contingent performance metrics

Participants are encouraged to define additional metrics relevant to their proposed solution.

Optional Tasks

A. Prototype

Create a prototype demonstrating key workflows or interactions of your proposed solution.

This may be a:

- Product mockup
- Dashboard
- Management portal
- Analytics platform

- Process visualization

B. Cultural Analytics Framework

Design a framework for collecting, tracking, and utilizing performance data across cultural sections.

C. Three-Year Roadmap

Develop a phased roadmap showing how your solution evolves over a three-year period.

D. Benchmarking Study

Compare IIT Roorkee's cultural ecosystem with practices adopted by leading institutes, organizations, sports teams, performing arts academies, or competitive student communities and identify actionable insights.

---

4. Deliverables

Deliverable 1: Strategy Document

Format: PDF

The document should include:

- Executive Summary
- Problem Analysis
- Root Cause Analysis
- Stakeholder Analysis
- Product Strategy
- Proposed Solution
- MVP/MVI Design
- Success Metrics
- Implementation Roadmap

Maximum Length: 15 Pages (excluding appendices)

Deliverable 2: Executive Pitch Deck

Format: PDF

Maximum Length: 10 Slides

The presentation should clearly communicate:

- Problem Statement
- Key Insights
- Proposed Solution
- Strategic Impact
- Success Metrics
- Implementation Plan

Deliverable 4 (Optional): Prototype

Format: Figma, Framer, Adobe XD, or equivalent

If submitted, the prototype should strengthen and support the proposed solution.

---

5. Guidelines

1. Focus on Diagnosis Before Solution

The strongest submissions will spend significant effort understanding the problem before proposing a solution.

A well-identified problem is often more valuable than a sophisticated solution.

---

2. Think Systemically

Avoid treating symptoms in isolation.

Consider the entire cultural talent lifecycle—from discovery and onboarding to training, performance, and long-term retention.

---

3. Prioritize Ruthlessly

A focused intervention addressing a critical bottleneck is often more impactful than a broad solution attempting to solve everything.

---

4. Support Assumptions with Reasoning

Where data is unavailable, clearly state assumptions and explain your reasoning.

---

5. Think Beyond Apps

Participants are not restricted to software products.

Processes, programs, frameworks, dashboards, platforms, and hybrid solutions are all acceptable if they create measurable value.

---

6. Design for Sustainability

The proposed solution should remain valuable across multiple years and batches of students.

---

7. Use Data to Drive Decisions

Participants are encouraged to define measurable success indicators and demonstrate how data can be leveraged to improve decision-making and outcomes over time.

Evaluation Criteria

Criterion	Weightage
Problem Discovery & Root Cause Analysis	25%

Product Thinking & Strategy	25%
Stakeholder Understanding	15%
Prioritization & Feasibility	15%
Innovation & Creativity	10%
Communication & Presentation	10%

Strategy & Management Consultancy

Market Entry Strategy for a Cultural Experiences Venture

Participation Format

This is a team problem statement.

Teams may consist of up to the maximum number of 02 participants.

Participants are expected to assume the role of a strategy consulting team advising a venture-backed company seeking to build a scalable business around India's traditional art and cultural ecosystem.

The challenge evaluates strategic thinking, market analysis, market sizing, business model design, go-to-market planning, financial reasoning, growth strategy, and communication skills.

---

1. Motivation

Across the world, consumers are increasingly seeking authentic experiences, community-driven activities, and culturally meaningful products. This trend has fueled the growth of businesses built around experiential entertainment, heritage tourism, creative communities, wellness experiences, and lifestyle brands.

India possesses one of the richest cultural ecosystems in the world, encompassing centuries-old traditions in dance, music, theatre, visual arts, storytelling, and craftsmanship. Yet many of these traditions remain fragmented, under-commercialized, and disconnected from mainstream consumer markets.

At the same time, rising disposable incomes, growing demand for experiential consumption, increasing interest in cultural tourism, and the emergence of creator-led economies present a significant opportunity to build scalable businesses around cultural experiences.

The challenge is not merely preserving cultural heritage.

The challenge is identifying how cultural heritage can be transformed into a commercially viable, scalable, and sustainable business while preserving its authenticity and long-term value.

2. Client Situation

A venture-backed lifestyle and entertainment company plans to launch a new business vertical focused on traditional Indian art and cultural experiences.

The company has secured seed funding and intends to launch within the next 12 months.

The leadership team believes there is a large untapped opportunity within India's cultural economy but lacks clarity on:

- Which art forms or cultural experiences to prioritize
- Which customer segments to target
- What business model to pursue
- How to position the venture
- How to launch and scale efficiently

You have been hired as a strategy consulting team to develop a market-entry strategy and growth blueprint for this new venture.

Your recommendations should help the company build a scalable, financially viable, and culturally responsible business.

3. Problem Statement

Develop a comprehensive market-entry strategy for a venture seeking to commercialize traditional Indian art and cultural experiences.

Your objective is to answer the following strategic questions.

A. Opportunity Selection

Which traditional Indian art, dance, music, theatre, craft, or cultural experience represents the most attractive business opportunity?

Your recommendation should be supported by factors such as:

- Cultural uniqueness
- Consumer demand
- Market size
- Commercial viability
- Scalability
- Availability of talent
- Competitive landscape
- Long-term growth potential

Teams may select one or more cultural experiences if appropriately justified.

B. Business Model Design

What is the most viable business model for the venture?

Potential approaches may include:

Experiences

- Cultural festivals
- Live performances
- Heritage walks
- Immersive exhibitions
- Experience centers

Learning & Education

- Workshops
- Certification programs
- Online courses
- School partnerships
- Corporate learning initiatives

Products & Commerce

- Artist-led merchandise
- Home décor
- Fashion collaborations
- Premium collectibles

Hybrid Models

A combination of multiple revenue streams.

Teams should justify why their proposed model offers the strongest commercial potential.

C. Customer Strategy

Who should be the primary target customer?

Develop detailed customer personas for key customer segments.

Potential segments include:

- Urban Millennials
- Gen Z Consumers
- Premium Travelers
- High-Net-Worth Individuals
- Educational Institutions
- Corporate HR & L&D Teams

For each segment, identify:

- Demographics
- Motivations
- Pain Points
- Purchasing Power
- Purchase Drivers
- Preferred Discovery Channels

D. Go-To-Market Strategy

Develop a launch strategy covering:

Positioning

How should the venture be positioned in the market?

Pricing

What pricing strategy should be adopted?

Distribution

How will customers access the offering?

Customer Acquisition

How will awareness and demand be generated?

Potential channels include:

- Influencer Marketing
- Creator Partnerships
- Performance Marketing
- Strategic Partnerships
- Campus Activations
- Experiential Marketing
- Community Building

Teams should clearly explain how their GTM strategy will drive awareness, acquisition, and retention.

E. Growth & Expansion Strategy

Develop a roadmap for scaling the venture.

Your recommendations should address:

- Geographic expansion
- Talent acquisition and onboarding
- Partnership strategy
- Growth levers
- Competitive differentiation

The objective is to demonstrate how the venture can achieve sustainable growth over time.

---

4. Mandatory Analysis

A. Market Sizing

Estimate:

- Total Addressable Market (TAM)
- Serviceable Addressable Market (SAM)
- Serviceable Obtainable Market (SOM)

All assumptions must be clearly stated and justified.

B. Competitive Landscape

Analyze:

- Existing competitors
- Substitute offerings
- Market gaps
- White-space opportunities

Teams may use frameworks such as:

- Porter's Five Forces
- SWOT Analysis
- Ansoff Matrix
- Value Chain Analysis

C. Business Model Evaluation

Compare at least two business model alternatives before recommending a final approach.

Clearly explain:

- Trade-offs
- Risks
- Revenue potential
- Scalability considerations

D. Financial Viability

Provide a high-level commercial assessment covering:

- Revenue streams
- Pricing assumptions
- Cost drivers
- Unit economics assumptions

Participants are not expected to build advanced valuation models.

E. Risk Assessment

Identify key risks such as:

- Consumer adoption risk
- Operational challenges
- Talent constraints
- Competitive pressures
- Cultural authenticity concerns

Recommend mitigation strategies for each identified risk.

---

5. Deliverables

Deliverable 1: Executive Summary

Format: PDF

Maximum Length: 1 Page

The summary should cover:

- Opportunity Selected
- Business Model
- Target Customer
- Go-To-Market Strategy
- Key Financial Insights
- Strategic Recommendation

The document should be suitable for review by senior leadership.

Deliverable 2: Consulting Presentation Deck

Format: PDF

Maximum Length: 15 Slides

The presentation should communicate:

- Market Opportunity
- Strategic Analysis
- Customer Insights
- Business Model
- Go-To-Market Strategy
- Financial Viability
- Growth Roadmap

The deck should be structured, concise, and boardroom-ready.

Deliverable 3: Market Sizing & Financial Model

Format:

- Microsoft Excel Workbook
- Google Sheets

The model should include:

- TAM/SAM/SOM Analysis
- Key Assumptions
- Revenue Estimates
- Sensitivity Analysis

The model should be transparent and easy to audit.

---

6. Guidelines

1. Think Like a Consultant

The strongest submissions will demonstrate structured problem-solving and evidence-based recommendations.

---

2. Prioritize Strategic Clarity

A focused and coherent strategy is often stronger than a collection of disconnected ideas.

---

3. Support Assumptions

Where data is unavailable, clearly state assumptions and explain the reasoning behind them.

---

4. Balance Culture and Commerce

Recommendations should preserve authenticity while ensuring commercial sustainability.

---

5. Focus on Execution

Strong strategies should be realistic, scalable, and implementable in the real world.

---

6. Communicate Clearly

The ability to communicate insights and recommendations effectively is as important as the analysis itself.

---

7. Evaluation Criteria

Criterion	Weightage
Strategic Thinking & Problem Structuring	25%
Market Analysis & Market Sizing	20%
Business Model Design	20%
Go-To-Market & Growth Strategy	15%
Financial Reasoning	10%
Communication & Storytelling	10%

Scaling Campus Threads – Growth Strategy for a National Student Commerce Brand

Participation Format

This is a team problem statement.

Teams may consist of up to the maximum number of 02 participants.

Participants are expected to assume the role of a strategy consulting team advising a high-growth startup seeking to scale beyond its initial market and establish itself as a national brand.

The challenge evaluates strategic thinking, growth strategy, market analysis, customer segmentation, go-to-market planning, unit economics, financial reasoning, and communication skills.

---

1. Motivation

India is home to over 40 million students across thousands of higher educational institutions, making it one of the largest youth markets in the world.

Over the last decade, student merchandise has evolved from simple event-specific apparel into a rapidly growing category encompassing:

- College Merchandise
- Club and Society Merchandise
- Alumni Collections
- Graduation Memorabilia
- Fest Merchandise
- Creator-Led Merchandise
- Personalized Campus Products

Students increasingly view merchandise as a symbol of identity, belonging, and community. Simultaneously, institutions, clubs, alumni associations, and student bodies have become significant consumers of customized products and experiences.

While several startups have emerged to serve this demand, scaling beyond a limited number of campuses remains challenging. Companies often struggle with customer acquisition, operational complexity, profitability, and differentiation.

The challenge is not acquiring the first few customers.

The challenge is building a scalable and defensible business capable of serving thousands of institutions while maintaining strong unit economics and sustainable growth.

---

2. Client Situation

Campus Threads is a venture-backed merchandise startup that has achieved product-market fit across 20 colleges and generated approximately ₹2 crore in annual revenue.

The company currently offers:

- Customized Apparel
- Club Merchandise
- Event Merchandise
- Graduation Merchandise
- Alumni Collections

After establishing a successful presence in its initial markets, the founders now aim to build India's largest student-focused commerce brand over the next five years.

However, several strategic questions remain unanswered:

- Which customer segments should be prioritized?
- What should be the company's primary growth engine?
- How should the company expand geographically?
- How can customer acquisition be scaled profitably?
- How can the company defend itself against increasing competition?

The founders have engaged your consulting team to develop a growth strategy that can transform Campus Threads from a niche merchandise startup into a nationally recognized student commerce platform.

---

3. Problem Statement

Develop a comprehensive growth strategy for Campus Threads.

Your objective is to identify how the company can achieve sustainable and profitable growth while building a long-term competitive advantage.

Your recommendations should address the following strategic questions.

A. Market Opportunity Assessment

Evaluate the attractiveness of the opportunity.

Your analysis should include:

- Total Addressable Market (TAM)
- Serviceable Addressable Market (SAM)
- Serviceable Obtainable Market (SOM)

Clearly state and justify all assumptions.

Participants should identify:

- Market size
- Growth drivers
- Emerging trends
- Attractive customer segments

B. Customer Segmentation & Prioritization

Identify and prioritize the most attractive customer segments.

Potential segments include:

- IITs
- NITs
- Central Universities
- Private Universities
- Colleges
- Student Clubs

- Fest Organizing Teams
- Alumni Associations
- Student Creators and Influencers

For each segment, evaluate:

- Market attractiveness
- Revenue potential
- Ease of acquisition
- Retention potential
- Strategic fit

Teams should recommend which segments Campus Threads should prioritize over the next three years.

C. Growth Engine Selection

Campus Threads can pursue multiple growth avenues.

Examples include:

Geographic Expansion

- New Cities
- New States
- Regional Clusters

Product Expansion

- Premium Merchandise
- Accessories
- Lifestyle Products
- Graduation Kits
- Collectibles

Customer Expansion

- Alumni Communities
- Schools
- Coaching Institutes
- Student Organizations

Platform Expansion

- Creator-Led Merchandise
- Marketplace Models
- Community Commerce

Participants should evaluate multiple growth paths and recommend the most attractive strategy.

D. Go-To-Market Strategy

Develop a customer acquisition strategy capable of scaling nationally.

Potential channels include:

- Campus Ambassador Programs
- Student Influencers
- Social Media Marketing
- Referral Programs
- Alumni Networks
- Partnerships with Student Bodies
- Community Building Initiatives

Teams should explain:

- Acquisition Strategy
- Conversion Strategy
- Retention Strategy
- Channel Prioritization

E. Competitive Advantage & Defensibility

As the market grows, competition is likely to increase.

Identify how Campus Threads can build a sustainable competitive moat.

Potential sources of advantage include:

- Brand
- Technology
- Personalization
- Fulfillment Network

- Community Effects
- Partnerships
- Data & Customer Insights

Participants should recommend how the company can remain differentiated over the long term.

F. Strategic Recommendation

Assume the founders can pursue only ONE primary growth strategy over the next three years.

Teams must recommend:

- The growth strategy they would prioritize
- Why it is superior to alternatives
- Expected business impact
- Key risks
- Implementation roadmap

This recommendation should serve as the central conclusion of the case.

---

4. Mandatory Analysis

A. Market Sizing

Estimate:

- TAM
- SAM
- SOM

Clearly explain assumptions and methodology.

B. Competitive Landscape

Analyze:

- Existing merchandise providers
- Local competitors
- Custom-printing platforms
- Emerging startups
- Alternative solutions

Teams may use frameworks such as:

- Porter's Five Forces
- SWOT Analysis
- Competitive Positioning Maps

C. Growth Option Evaluation

Compare at least three growth alternatives.

Examples:

- Geographic Expansion
- Product Expansion
- Customer Segment Expansion
- Platform Expansion

Clearly explain:

- Benefits
- Risks
- Scalability
- Financial implications

D. Unit Economics Assessment

Estimate and discuss:

- Customer Acquisition Cost (CAC)
- Lifetime Value (LTV)
- Gross Margins
- Contribution Margins

Discuss how the proposed strategy impacts long-term profitability and sustainability.

E. Risk Assessment

Identify key risks including:

- Competitive pressure
- Operational complexity
- Demand fluctuations
- Customer concentration
- Supply chain challenges
- Brand dilution

Recommend mitigation strategies.

---

5. Deliverables

Deliverable 1: Executive Summary

Format: PDF

Maximum Length: 1 Page

The summary should include:

- Growth Recommendation
- Target Customer Segments
- Go-To-Market Strategy
- Unit Economics Highlights
- Strategic Rationale

The document should be suitable for review by founders and investors.

Deliverable 2: Consulting Presentation Deck

Format: PDF

Maximum Length: 15 Slides

The presentation should communicate:

- Market Opportunity
- Customer Insights
- Growth Strategy
- Business Model
- Competitive Analysis
- Unit Economics
- Expansion Roadmap
- Final Recommendation

The deck should be concise, structured, and boardroom-ready.

Deliverable 3: Market Sizing & Financial Model

Format:

- Microsoft Excel Workbook
- Google Sheets

The model should include:

- TAM/SAM/SOM Analysis
- Revenue Projections
- Key Assumptions
- Unit Economics
- Sensitivity Analysis

The model should be transparent and easy to audit.

---

6. Guidelines

1. Think Like Growth Consultants

The strongest submissions will focus on scalable growth rather than isolated tactical recommendations.

---

2. Support Recommendations with Analysis

Every recommendation should be backed by quantitative reasoning and structured analysis.

---

3. Prioritize Focus

A focused growth strategy is often more effective than pursuing multiple opportunities simultaneously.

---

4. Balance Growth and Profitability

Recommendations should improve both scale and long-term economic sustainability.

---

5. Consider Execution Challenges

Strategies should be practical, realistic, and implementable.

---

6. Defend Trade-Offs

There is no single correct answer. Strong teams will demonstrate why their chosen path is superior to competing alternatives.

7. Evaluation Criteria

Criterion	Weightage
Strategic Thinking & Growth Strategy	25%
Market Analysis & Market Sizing	20%
Customer Segmentation & GTM Strategy	15%
Competitive Advantage & Business Model	15%
Unit Economics & Financial Reasoning	15%
Communication & Storytelling	10%

Graphic Design

Designing a Modern Visual Identity for the Cultural Council

Participation Format

This is an individual problem statement.

Participants are required to work and submit individually. All concepts, visual assets, mockups, presentations, and supporting materials must be created by a single participant.

The challenge evaluates brand strategy, visual design, design systems thinking, visual communication, and the ability to create cohesive brand experiences across digital and physical touchpoints.

---

1. Motivation

The Cultural Council of IIT Roorkee serves as the creative nucleus of the institute, bringing together over twenty clubs and sections spanning dramatics, dance, music, cinema, literature, photography, design, and other cultural disciplines.

Through performances, productions, workshops, competitions, and festivals, the Council shapes a significant part of the student experience and campus culture.

Despite its impact and visibility, the Council's visual identity lacks a unified and recognizable brand system. In today's design landscape, successful organizations are identified not merely by a logo, but by a consistent visual language that communicates their values, personality, and aspirations across every interaction.

A strong brand identity improves recognition, strengthens communication, and creates a lasting impression among students, alumni, visitors, and external stakeholders.

This challenge invites participants to reimagine the Cultural Council's visual identity and create a modern, cohesive, and scalable brand system that reflects the creativity, diversity, collaboration, and cultural excellence of IIT Roorkee.

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2. Problem Statement

Assume the role of a Brand Designer tasked with redesigning the visual identity of the IIT Roorkee Cultural Council.

Your objective is to create a modern and cohesive visual identity that can be consistently applied across digital platforms, social media, merchandise, event branding, and official communications.

The final solution should demonstrate not only visual appeal but also strong design thinking, consistency, scalability, and practical usability.

Participants are encouraged to build a brand that:

- Reflects the Council's cultural diversity.
- Appeals to current and prospective students.
- Feels modern and future-ready.
- Maintains consistency across applications.
- Creates a recognizable identity for the Cultural Council.

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3. Design Requirements

Mandatory Components

A. Logo System

Design a visual identity system consisting of:

- Primary Logo
- Secondary or Alternate Logo
- Monochrome Version
- Social Media Profile Mark / Icon

The logo should remain recognizable and effective across different sizes and mediums.

B. Typography System

Define:

- Primary Typeface
- Secondary Typeface

Participants should justify their typography choices and explain how they align with the overall brand personality.

C. Color Palette

Develop a cohesive color system including:

- Primary Colors
- Secondary Colors
- Accent Colors

Color choices should support both digital and physical applications.

Participants should provide:

- HEX Codes
- RGB Values

D. Social Media Brand Application

Design a set of social media templates demonstrating how the identity would be used online.

Required:

- Announcement Post
- Event Promotion Post
- General Communication Post

These may be presented as a three-post Instagram grid.

E. Physical Brand Application

Demonstrate the identity on any ONE of the following:

- Hoodie
- T-Shirt
- Lanyard
- Event Banner
- ID Card
- Tote Bag

The objective is to show how the brand translates beyond screens.

Optional Components

A. Motion Identity

Create a short animated logo reveal.

Recommended Duration:

- 3–5 Seconds

B. Section-Level Extensions

Demonstrate how the identity could be adapted for one or more Cultural Council sections while maintaining consistency with the parent brand.

Examples:

- Dramatics Section
- Cinematic Section
- Choreography Section
- Literary Section
- Audio Section

C. Additional Brand Applications

Examples:

- Website Landing Page

- Event Pass
- Sticker Pack
- Merchandise Collection
- Digital Signage

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4. Deliverables

Deliverable 1: Design Assets

Submit:

- Logo Files (PNG and SVG)
- Color Specifications
- Typography Specifications
- Social Media Designs

All assets should be organized and clearly labeled.

Deliverable 2: Mockups

Submit high-quality mockups showcasing the brand in real-world applications.

At least one physical application is mandatory.

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5. Guidelines

1. Design Beyond the Logo

The strongest submissions will focus on building a complete visual identity rather than designing a standalone logo.

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2. Prioritize Consistency

Every design element should feel like part of a unified system.

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3. Focus on Practicality

The identity should work effectively across both digital and physical mediums.

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4. Communicate Design Decisions

Strong visual outcomes should be supported by clear reasoning and design rationale.

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5. Maintain Originality

All submitted work must be original.

Direct imitation or close resemblance to existing brands is discouraged.

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6. Evaluation Criteria

Criterion	Weightage
Brand Strategy & Design Thinking	25%
Logo & Visual Identity Design	25%
Consistency of Design System	20%
Digital & Physical Applications	15%

Creativity & Originality	10%
Presentation & Communication	5%

Bonus Considerations

Additional credit may be awarded for:

- Motion Identity
- Section-Level Brand Extensions
- Additional Brand Applications
- Exceptional Visual Storytelling
- Innovative Design Systems
